From Signal to Strategy: The Development Story

Why each of the 8 PEAD strategies was built, in the order it was built
Companion to *PEAD_Report.pdf* (evidence the effect exists) and *PEAD_Strategy_Showcase.pdf* (full performance numbers)

PEAD_Report.pdf establishes that stocks with the most positive earnings surprises keep drifting up, and stocks with the most negative surprises keep drifting down, for weeks after the announcement. Knowing a pattern exists is not the same as knowing how to trade it well. This report walks through, in order, how a simple first attempt at trading that pattern was tested, found wanting in a specific and measurable way, and improved — eight times — into the strategies compared in *PEAD_Strategy_Showcase.pdf*. Two dead ends are included as well, because ruling something out is also a real result.

A few terms used throughout this report, in plain language:
- **Long/short:** buying stocks you expect to go up (the "long" side) while simultaneously selling borrowed stocks you expect to go down (the "short" side), so the strategy can profit whether the whole market rises or falls.
- **Dollar-neutral:** the long side and the short side are sized to the same total dollar amount, so a plain market-wide move up or down affects both sides roughly equally and cancels out.
- **Decile:** splitting all of a quarter's earnings events into 10 equal-sized groups, ranked from most negative surprise (decile 1) to most positive (decile 10).
- **Leverage cap:** a limit on the total dollar size of all positions combined, relative to the strategy's own capital, so it can't take on unlimited risk.
- **Liquidity cap:** a limit on any single position's size relative to how much of that stock actually trades in a day, so the backtest never assumes a trade the real market couldn't absorb.
- **Sector/size-neutral:** spreading a strategy's weight evenly across industries (or firm sizes) so its return isn't secretly driven by one industry or size bucket outperforming for reasons unrelated to earnings surprises.
- **Sharpe ratio:** return earned per unit of volatility (risk) taken on -- higher is better risk-adjusted performance, independent of how large the raw return is.

Stage 1: three first attempts (v1)

The first version of this project asked the most direct question possible: can the decile-drift pattern be turned into positions and traded? Before building any strategy, one basic question had to be answered first: **how long should a position actually be held?** Rather than guess, the size+book-to-market-adjusted return was tracked out to 80, 100, 120, 150, and 180 trading days, and each decile's *marginal* return -- the extra return earned specifically between two adjacent checkpoints -- was tracked to find where it flattens toward zero.

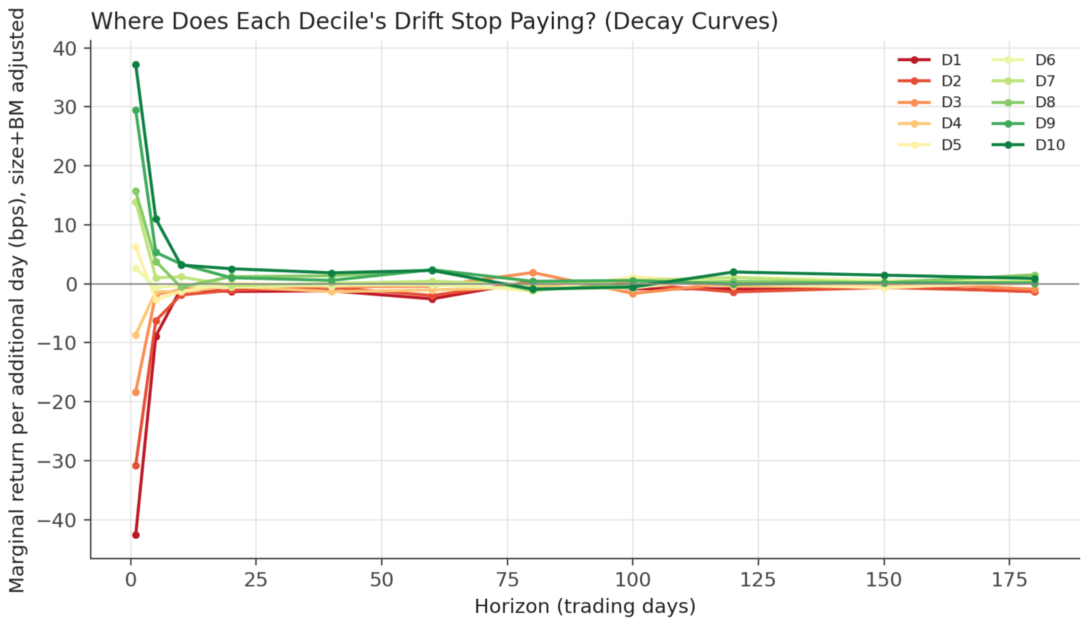

Figure 1. Marginal return per additional trading day of holding, by decile, size+book-to-market adjusted. Every decile's marginal contribution is large and directional in the first 5-10 days, then settles toward a small, noisy band close to zero by roughly day 20-40. The extreme deciles (D1, D2, D9, D10) keep contributing a small but consistently signed amount out to 80 days; the near-median deciles (D5, D6) flatten out almost immediately, consistent with their weak, often statistically insignificant drift documented in *PEAD_Report.pdf*.

From this, three holding-period buckets are used across every strategy in this project: **extreme** deciles (1, 2, 9, 10) hold 80 trading days; **shoulder** deciles (3, 4, 7, 8) hold 40 trading days; **near-median** deciles (5, 6) hold 20 trading days, a short duration reflecting that there is little measured edge there to wait for. This is a deliberate simplification of a noisier per-decile decay estimate into three defensible, coarser bands rather than chasing every noisy inflection point.

With holding periods settled, three portfolio constructions were built and compared side by side, all using a fixed-notional backtest (positions sized off a fixed starting capital, not a compounding balance):

- **Extreme decile long/short** — long every D10 event, short every D1 event, the simplest possible framing.
- **Rank-weighted** — every decile traded, with a position weight that scales continuously with how extreme the surprise rank was, rather than throwing away 80% of the cross-section at a hard cutoff.
- **Balanced** — the extreme-decile trade, but with each leg rebalanced across firm size quintiles so neither leg is secretly dominated by one size bucket.

All three were profitable net of transaction costs and liquidity limits. Rank-weighted had the better Sharpe ratio (it diversifies across far more positions), but the lower raw return of the three — a tradeoff, not a clean win, and the first real question the project needed to answer.

Stage 2: why did rank-weighted trade Sharpe for return? -> Strategy 4

Rather than accept the tradeoff, the project measured *why* it existed. Rank-weighted trades roughly 4x as many positions as the extreme-decile strategies (it holds a small position in every decile, not just the top and bottom). That volume ran the strategy into two real capacity constraints: the **liquidity cap bound on 99% of its positions** (nearly every position wanted to be larger than the 1% of daily trading volume it was allowed), and the **leverage cap bound in 70% of its quarters**. Worse, the old leverage cap trimmed every position down by the same percentage when it bound — shaving the strategy's highest-conviction bets (the most extreme surprises) by exactly as much as its weakest, lowest-conviction ones.

Strategy 4 addressed this directly, not by tuning a parameter but by changing the construction: **trim** the low-conviction middle of the SUE-rank distribution out entirely (don't even hold small positions in near-median events), **tilt** remaining weight toward the extremes, keep size-neutrality, and replace proportional cap-trimming with **priority-based** trimming — when the leverage cap binds, the lowest-conviction positions still in the book are dropped first, protecting the strongest convictions.

Result: the first strategy to improve *both* return and Sharpe over the v1 baseline at the same time, rather than trading one for the other.

Stage 3: an unpriced risk hiding in Strategy 4 -> Strategy 5

With Strategy 4 working, the project ran a diagnostic most backtests skip: does the book's industry composition drift away from the overall market's, in a way that has nothing to do with earnings surprises? It did. In some quarters, Strategy 4's book had up to **29% of its total gross weight concentrated in a single Fama-French industry sector**, versus roughly 8.3% if weight were spread evenly across all 12 sectors.

That concentration is a real, uncompensated risk: if that one sector has a bad quarter for reasons unrelated to earnings surprises, the strategy takes the hit regardless of whether its PEAD signal was right. **Strategy 5** adds sector-neutrality on top of the existing size-neutrality — redistributing weight within each sector the same way it was already being redistributed within each size quintile.

Result: Strategy 5 has a *lower* raw return than Strategy 4 (4.06% vs. 5.26%), but the **best Sharpe ratio (1.33) and shallowest max drawdown (-10.8%) of all 8 strategies built**. This is the clearest example in the whole project of a decision that traded some return for a larger reduction in un-earned risk — a good trade on a risk-adjusted basis, and it became the foundation every later strategy builds on.

Stage 4: was the leverage cap still doing anything useful? -> Strategy 6

Strategy 5's leverage cap (1.5x gross exposure relative to capital) had originally been set as a conservative guard against exactly the kind of concentration risk Stage 3 just removed. With that risk gone, was the cap still protecting against anything, or just leaving return on the table? The project swept the cap from 1.5x up through 10x and measured what happened to Sharpe and return at each level.

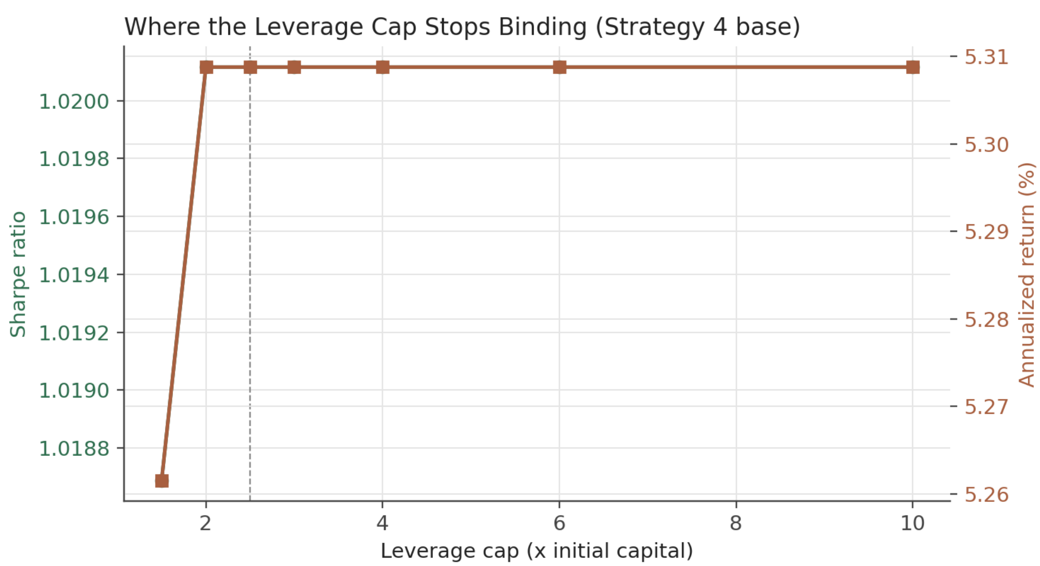

Figure 1. Both Sharpe ratio and annualized return flatten out completely by around 2.0-2.5x leverage (dashed line) -- past that point, raising the cap further changes nothing, because the strategy's own position-level and liquidity limits are what's actually constraining it, not the leverage cap. Below that point, the cap was costing real return for no risk-reduction benefit.

Strategy 6 raises the leverage cap to 2.5x, the point where it stops binding entirely. Result: both return (5.08% -> 6.49%, at the same base position-sizing parameter) and Sharpe improved versus Strategy 5, because there was no concentration risk left for the tighter cap to be protecting against. Strategy 6 is the strategy this project recommends as the pure-alpha core for live consideration.

Stage 5: Strategy 6 still trails the S&P 500 -- is that a problem?

Strategy 6 earns 6.49% a year. The S&P 500 earned roughly double that over the same window. Before treating that as a shortfall, it's worth being precise about what's being compared. Every strategy in this project is deliberately built market-adjusted and dollar-neutral, which means it structurally gives up the equity risk premium that makes up most of a simple buy-and-hold return — by design, not by accident. As an external sanity check, AQR's live Equity Market Neutral Fund (QMNNX) returns about 6.19% a year at similar volatility and Sharpe (0.69), and it *also* trails the S&P 500 by a wide margin — that is what a professionally run market-neutral strategy is supposed to look like. Strategy 6 already beats that real fund's Sharpe (1.30 vs. 0.69).

Still, the project treated "can we responsibly close some of that gap" as a real research question, and tested two different ways to do it, guided by how institutions actually approach this in practice: blend the alpha into a multi-factor book and add market exposure back as a *separate* position on top of an untouched alpha book (a "130/30"-style structure), rather than distorting the alpha signal itself to let beta leak in.

6a. Strategy 6 + 0.5x market beta overlay -- the recommended way

Strategy 6's alpha book is left completely untouched. A separate, synthetic market-index position sized to 0.5x the strategy's own trailing NAV is held alongside it, rebalanced on the same quarterly cadence with its own small transaction cost. Result: **10.08% return**, squarely inside a 10-15% equity-like target band, with a meaningfully shallower drawdown and better Sharpe than simply holding the market outright over the same period.

6b. Strategy 7 -- the other way, and why it's a cautionary result

The alternative way to let some market exposure through is to stop forcing the long leg's and short leg's dollar totals to match every quarter (Strategy 5/6's neutralization does this by construction). Removing that constraint required switching from market-adjusted to **raw** returns, because market-adjusting would have silently cancelled out the very net exposure this test was trying to let through.

The result is the most important lesson of this whole project: despite averaging only -5.5% net exposure as a fraction of gross (essentially flat, by that one measure), Strategy 7's realized correlation to the market came out at **0.52** — far higher than the small, controlled net-dollar tilt the strategy appeared to be taking. **Dollar-neutral is not the same thing as beta-neutral.** Positive- and negative-surprise firms structurally differ in average market beta, so holding equal dollar amounts long and short doesn't cancel market risk the way market-adjusting the underlying returns does. Strategy 7 is kept in the repository, fully documented and runnable, specifically because it demonstrates this distinction — not because it is a recommended design.

Stage 6: how far can each design be pushed for more return? -> Aggressive variants

Every strategy above was tuned toward the best risk-adjusted outcome, not the highest return achievable. A separate question, asked directly rather than inferred: if an investor is willing to give up some Sharpe ratio for meaningfully more absolute annual return, what does that look like *without changing what the strategy actually believes* -- same signal, same trim, same tilt, same size/sector neutrality? Only the two dials that control how big and how levered the book is run -- base_unit_fraction and the gross-leverage cap -- were swept, for each of the 8 designs, up to a realistic 6x institutional leverage ceiling, with the real 5%-of-ADV liquidity cap left untouched throughout as the genuine capacity constraint it is.

The diagnostic finding here mirrors Stage 2's rank-weighted lesson from a different angle. Just as rank-weighted's extra positions ran into liquidity/leverage caps before adding real return, sizing any of these 8 designs up runs into the SAME caps faster than naive scaling would suggest: Strategy 6's own sizing sweep (see *PEAD_Strategy_Showcase.pdf* Figure 6) shows annualized return actually *peaking* around 4x its baseline sizing and declining beyond that point, as an ever-larger share of positions get liquidity-capped. The binding constraint chosen here was a **max drawdown floor of -40%** (a deeper drawdown being a fund-ending event, not a tunable risk preference) alongside a Sharpe floor of 0.6 -- and for every design except the beta overlay, the drawdown floor binds *before* Sharpe actually reaches 0.6, landing every aggressive variant's Sharpe in a 0.72-1.09 band while roughly doubling its own baseline's return.

A finding worth calling out on its own: Strategy 5 Aggressive and Strategy 6 Aggressive converge to the identical configuration. The two share the exact same weight construction -- Strategy 6's only difference from Strategy 5 was a higher starting leverage cap and sizing (Stage 4 above). Once both are allowed to search up to the same 6x ceiling, Strategy 6's original advantage disappears entirely: they land on the same base_unit_fraction and the same result. This is a clean confirmation, from data rather than assumption, that leverage cap and position size are really one underlying dial, not two independent ones -- exactly the kind of result this project's diagnostic approach (test the actual mechanism, don't assume it) was built to surface.

Full methodology, the complete sizing-sweep chart, per-strategy numbers, and the caveats that matter more at this sizing (liquidity-cap bite roughly quadruples; drawdowns deepen to -33% to -40%) are in *PEAD_Strategy_Showcase.pdf*'s "Aggressive variants" section.

Two deliberate dead ends

Not every research thread in this project improved the strategy, and both of the following are documented rather than quietly dropped, because ruling something out is itself useful information for anyone continuing this work.

Dead end 1: the announcement-day return doesn't add information

Question: does a firm's own price reaction on the announcement day itself (its "announcement-window return," or EAR) predict anything *beyond* what the earnings-surprise number (SUE) already predicts? An early version of this test found a strong, ever-increasing Sharpe as more weight was put on EAR — which turned out to be a bug: it used the return from day 0 to day 0+1, which mechanically overlaps with a position's own first day of held profit, since positions start accruing the day *after* entry. Fixed to use only day 0's own, already-realized return, the result flattens out entirely: Sharpe stays in a narrow ~1.00-1.06 band regardless of how much weight is put on EAR. Not incorporated into any numbered strategy.

Dead end 2: earnings surprises don't reverse the way the literature says they should

A well-known finding (Bernard & Thomas, 1990) is that a firm's own earnings surprise tends to *reverse* about four fiscal quarters later (they report roughly -0.24 autocorrelation) — a firm that beat big this quarter tends to miss next-year's comparable quarter, and vice versa. If true in this data, it would suggest a tradeable "fade your own past surprise" signal.

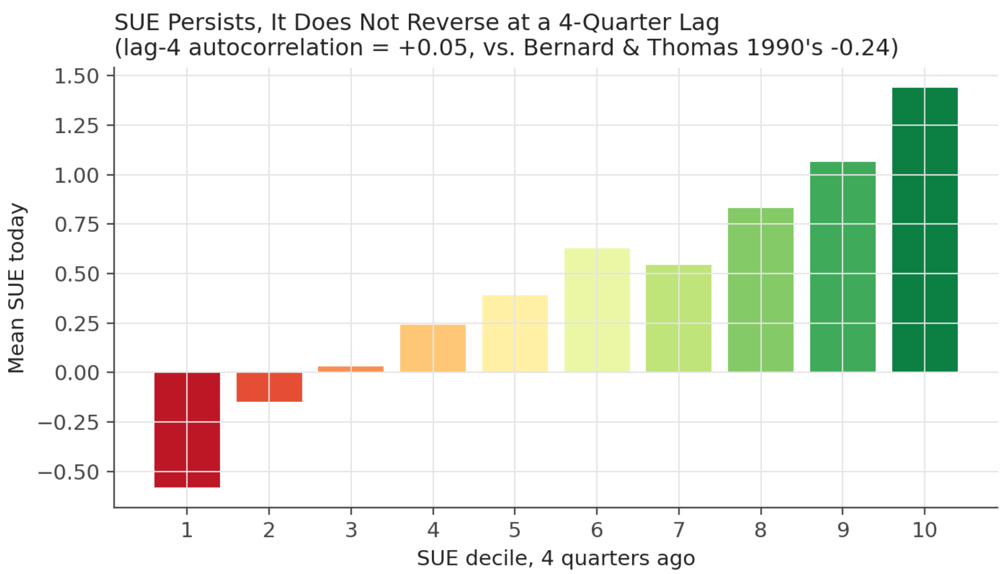

Figure 2. Average SUE today, grouped by each firm's own SUE decile four quarters earlier. If reversal were present, this line would slope *downward* (high past SUE predicting low current SUE). It slopes upward instead: firms with the highest surprise four quarters ago still average the highest surprise now.

In this data, the lag-4 autocorrelation is **+0.05** — the opposite sign from the literature, and the decile breakdown confirms it's not noise: it's clean **persistence**, not reversal. This is why none of Strategies 4-7 attempt to fade a firm's own earnings-surprise history.

The whole arc, in order

Stage	What changed	Why
1. v1 baseline	3 first-pass constructions built and compared	Establish that the signal is tradeable at all
2. Strategy 4	Trim + tilt + priority-based cap trimming	Rank-weighted saturated liquidity/leverage caps; proportional trimming diluted conviction
3. Strategy 5	+ Sector-neutral	Found up to 29% gross weight in one sector; an uncompensated risk
4. Strategy 6	Leverage cap 1.5x -> 2.5x	Swept the cap; found it stopped binding by ~2.0-2.5x once concentration risk was gone
5a. Strategy 6+beta	+ 0.5x market beta overlay (alpha book untouched)	Institutional-style way to close the return gap to equity-like targets
5b. Strategy 7	Removed long=short dollar constraint (raw returns)	The other way to add exposure -- found to leak far more beta than intended (cautionary)
Dead end	EAR signal blend	Caught its own look-ahead bug; flat Sharpe once fixed -- not adopted
Dead end	SUE lag-4 reversal test	Data shows persistence (+0.05), not reversal -- not adopted
6. Aggressive variants	base_unit_fraction/leverage cap pushed to a 6x ceiling, all 8 designs	Return roughly doubles per design; a -40% drawdown floor binds before Sharpe reaches 0.6

Where this leaves the project

Strategy 6 (pure alpha) and Strategy 6 + 0.5x beta overlay are the two designs recommended for live consideration, for the reasons laid out in Stage 5; Strategy 6 Aggressive and Strategy 6+beta Aggressive are the corresponding picks for a mandate that can tolerate a -35% to -40% drawdown for roughly double the return (Stage 6). Full performance numbers for every strategy and its aggressive variant are in *PEAD_Strategy_Showcase.pdf*; the underlying evidence that the earnings-drift effect being traded is real, and not an artifact of risk exposure or a narrow subsample, is in *PEAD_Report.pdf*.

Sources: README.md's documented strategy evolution, data/sweep_strategy4_leverage.csv, data/sue_reversal_summary.json, data/sue_reversal_by_decile.csv, data/sweep_aggressive_*.csv, and the sector-concentration and EAR diagnostics printed by scripts/equity_strategy/21_leverage_and_improvements.py, scripts/equity_strategy/22_sector_neutral_and_cadence.py, and scripts/equity_strategy/27_ear_signal_diagnostic.py. Scripts: scripts/equity_strategy/20_strategy4_tilted.py through scripts/equity_strategy/28_sue_reversal_test.py, scripts/equity_strategy/29_v2_strategy_charts.py, scripts/equity_strategy/31_build_development_report.py, scripts/equity_strategy/36_aggressive_variants.py, scripts/equity_strategy/37_aggressive_beta_overlay.py.